

Maintenance sheet version 10 for ablation area (tripod, one level)

Station name	NUK_B
People present	PHO, Thomas Varming, Geir (pilot)
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	02/10/23

Notes:

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced (Y/N)	Number shipped/arrival	Number departure	
Radiometer (write down the serial number you use!)	Y	Y	S: 212889/150711 150711		Clean? Y / N
Ablation hose assembly	N	N			No air in hose and bladder half full? Y / N
Iridium modem IMEI number	In loggerbox	Y	S: 300534063819850 checked		
Inclinometer/compass	In CNR4	Y	Attached to the radiometer? Y / N		
Satellite antenna	Y	Y			
Wind sensor	Y	Y	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	Y	Y	Fan spins? Y / N Sounds, OK? Y / N NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
HygroClip T/hum assembly	N	N			
Sonic ranger on AWS	Y	Y	Old sensor with new membrane? Y / N Unsure		
Sonic ranger on stake(s)	N	N	Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	N	N			
Solar panel	Y	Y	Clean? Y / N NB: is not effective when partially covered		
GPS antenna	In loggerbox	Y	At top of logger enclosure? Y / N		
Logger box ID	T3B-0006	Y	CR1000 or CR1000X? ID number: 41257		
Multiplexer	N	N	Clicking during measurement cycle? Y / N		
Logger enclosure	Y	Y	Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y	Y	New box? Y / N New desiccant bags? Y / N		
Stakes	N	N	New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	Y	Y	New tripod? Y / N New sensor boom? Y / N		

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Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	0m 0.35s	0m 0.35s
Download data and/or change CF/SD Card (wait for green light)	Y/N	Y/N
Name of logger program		PROMICE v3 - Rewired
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	°
Mast tilt across boom looking from radiometer (- if clockwise)	°	°
Boom direction relative to north	° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y/N	Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	Y/N	Y/N
Temperature / humidity height (bottom of casing - surface)	cm	cm
Stake assembly standing?	Y/N / sort of	Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	cm
Vertical difference in surface height (at station - at hose) (approx.)	cm	cm
Length of thermistor string on surface from surface marking	cm	cm
Apparent damage to sensors? If yes, which?	Y/N	Y/N

Notes: AWS SR50 height to surface: 2.27cm

~~Ensure if Tx Message test was successful. CHECK.~~

- The AWS SR50 seems to be off from the true measurement. SR50 says 1.80m, we measure it as 2.27m. We needed to rewire it to get it to work, specifically the green wire was in C4 and we moved it to C3 (to join the orange cable)
- AWS tripod stabilised with rocks.
- No dessicant in logger box or battery box
- Battery box secured to station with helicopter cargo strap

Must-do at next visit:

- Lufft sensor had to be mounted on top of station pole because we needed to use the bolt in the tripod fitting - the tripod did not have a bolt in the fitting to secure the height of the pole
- Iridium and GPS connection in logger box is loose, Tape used to secure it.
- Bring new SR50, bring mounting for lufft sensor and install correctly, Secure AWS to rock with rock bolts and drill; bring dessicant